Comparative Transcriptomic Analysis of Salt-Stress Responses in Barley

An exploratory RNA-seq workflow comparing wild/tolerant and cultivated/sensitive barley under control and salt-treated conditions.

Project overview

This project reanalysed publicly available barley RNA-seq data to explore transcriptional responses to salt stress in wild barley (Hordeum spontaneum, described in the study metadata as the tolerant genotype) and cultivated barley (Hordeum vulgare, described as the sensitive genotype). The workflow covered read retrieval and organisation, raw-read quality assessment, preprocessing, transcript quantification, sample-level quality assessment, differential expression analysis, and Gene Ontology (GO) enrichment.

Project objectives

- Build a reproducible end-to-end RNA-seq workflow using public sequencing data.
- Compare salt-treated and control samples within each genotype.
- Compare the genotypes under control and salt-treated conditions.
- Identify exploratory differentially expressed gene sets and functional categories for follow-up.

Scope and interpretation

The analysis is exploratory. Each group contains two biological replicates, and a potential library-type inconsistency was detected during Salmon diagnostics. Results should be treated as candidate signals rather than definitive biological conclusions.

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Tools: SRA Toolkit, FastQC, MultiQC, fastp, Salmon, R, tximport, and DESeq2.

Biological Question, Design and Metadata

The four-group design supports within-genotype treatment contrasts and between-genotype comparisons.

Research question

How do transcriptional profiles differ between wild/tolerant and cultivated/sensitive barley under control and salt-treated conditions, and which genes or GO categories may be associated with these differences?

Experimental groups

Group	Interpretation	Run accessions	n
TC	Wild/tolerant genotype, control	SRR16989368, SRR16989369	2
TS	Wild/tolerant genotype, salt-treated	SRR16989370, SRR16989371	2
SC	Cultivated/sensitive genotype, control	SRR16989364, SRR16989365	2
SS	Cultivated/sensitive genotype, salt-treated	SRR16989366, SRR16989367	2

Planned comparisons

Contrast	Question addressed
TS_vs_TC	Salt response in the wild/tolerant genotype
SS_vs_SC	Salt response in the cultivated/sensitive genotype
TC_vs_SC	Genotype difference under control conditions
TS_vs_SS	Genotype difference under salt treatment
Genotype × Treatment	Whether the treatment response differs between genotypes

Metadata checks

The curated sample_metadata.csv assigns all eight accessions to the four groups above. The SRA metadata also identifies the runs as paired-end and includes run, BioSample, experiment, genotype, treatment, and group fields. The SRA organism field lists Hordeum vulgare for all records, while the curated metadata notes that this may differ from the genotype descriptions in the publication. This is retained as a metadata caveat rather than silently resolved.

Data Retrieval and Quality Control

Public reads were organised and evaluated before quantification.

Data retrieval and file organisation

Eight paired-end SRA runs (SRR16989364–SRR16989371) were retrieved and converted to FASTQ using the SRA Toolkit. FASTQ files were organised under 01_FASTQ, raw-read QC reports under 03_QC_Raw, and processed reads under 04_Preprocessing. SRA files were removed after FASTQ files were verified, to conserve local storage.

Raw-read quality assessment

FastQC was run on raw reads, and MultiQC consolidated the reports. Review focused on per-base quality, adapter content, sequence duplication, GC distribution, and read-length patterns. The libraries were paired-end, and the reported read length was consistent with 151-bp reads.

Preprocessing

fastp was used to trim adapter contamination and perform read filtering. A representative run indicated that approximately 0.47% of reads were filtered, while quality scores were generally strong and Q30 metrics were good. Post-trimming FastQC and MultiQC reports were generated.

QC observations

- Insert-size distribution showed a peak around 184 bp in the reviewed report.
- GC profiles were broadly similar across samples.
- Sequence duplication was around 30%; this was noted but not used alone to discard samples.
- Adapter trimming was followed by post-trimming quality assessment.

Figure 1: MultiQC overview for raw reads (quality, GC, duplication and adapter content)

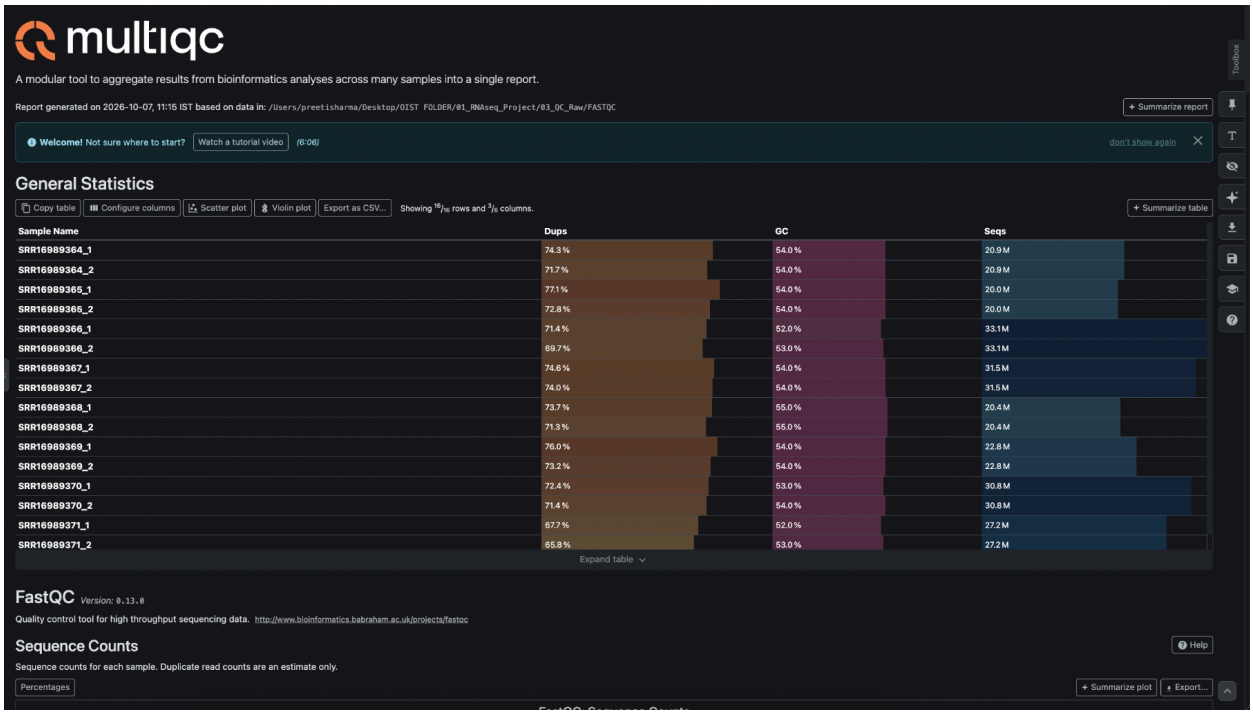
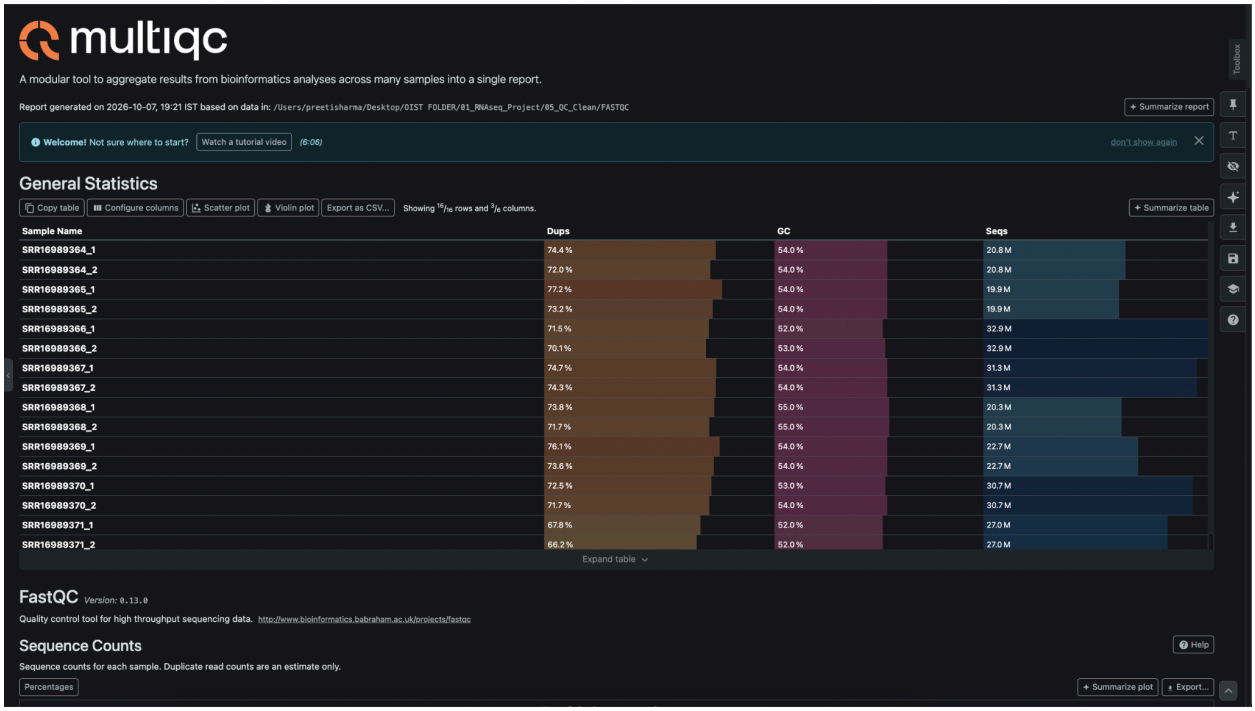


Figure 2. MultiQC summary of trimmed-read quality control.



Transcript Quantification and Sample-Level Assessment

Salmon and R-based tools estimated abundance and inspected sample relationships.

Reference and quantification

Transcript abundance was estimated with Salmon v2.8.0 using the Hordeum vulgare IBSC_v2 transcript reference. Reference files included Hordeum_vulgare.IBSC_v2.cdna.all.fa.gz, Hordeum_vulgare.IBSC_v2.42.gff3.gz, and a transcript-to-gene mapping table containing 248,391 mappings. The Salmon index contained 246,184 valid references.

Salmon quantification was completed for all eight samples using trimmed paired-end FASTQ files. Quantification outputs, including quant.sf files, were retained in sample-specific folders under 07_Quantification. Expression estimates were imported into R using tximport, with DESeq2 used downstream.

Sample-level QC

PCA and sample clustering were generated from imported expression data. The first two principal components explained 58% and 22% of variance, respectively (80% combined). Replicates generally clustered close to their corresponding group, and group-level separation was visible. PCA separation alone does not establish that a pattern is biological; technical effects and the small replicate count remain important.

Figure 3. Principal component analysis (PCA) of the eight barley RNA-seq samples based on gene-expression profiles. Each point represents an individual sample, coloured according to its experimental group (SC, SS, TC, or TS). PC1 and PC2 explain 58% and 22% of the total variance, respectively, accounting for 80% combined variance. The distribution of samples illustrates differences in expression profiles among the four experimental groups.

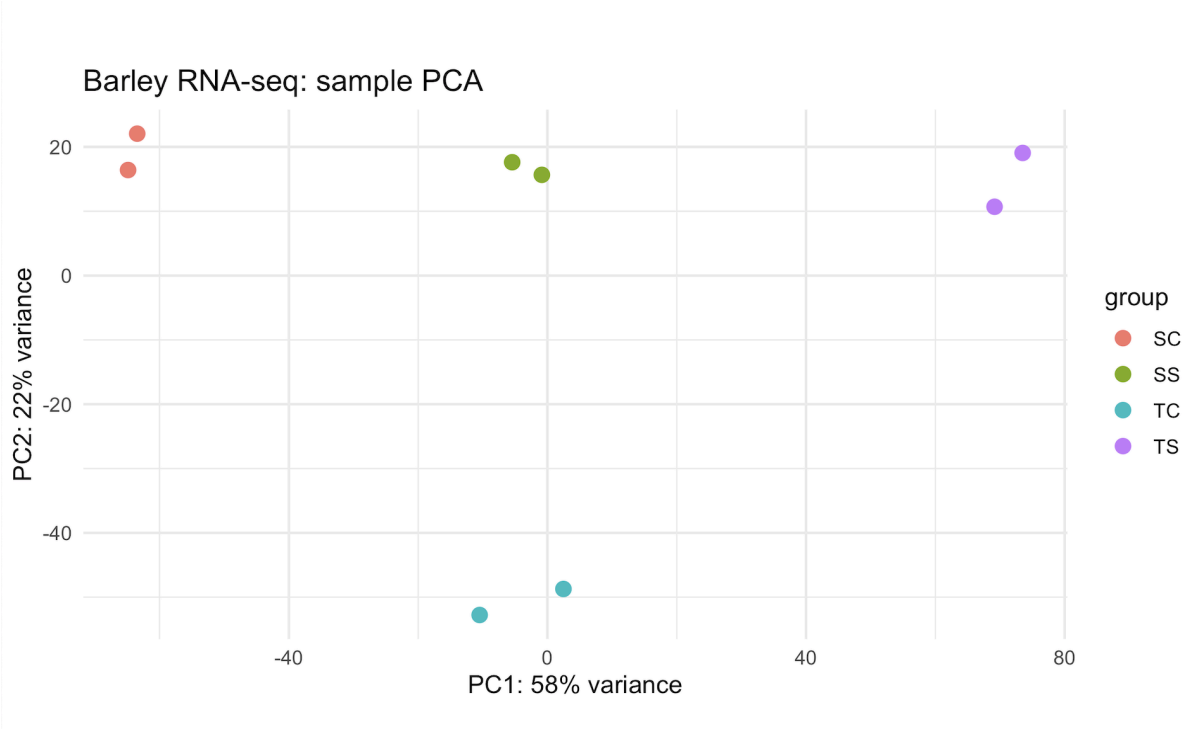
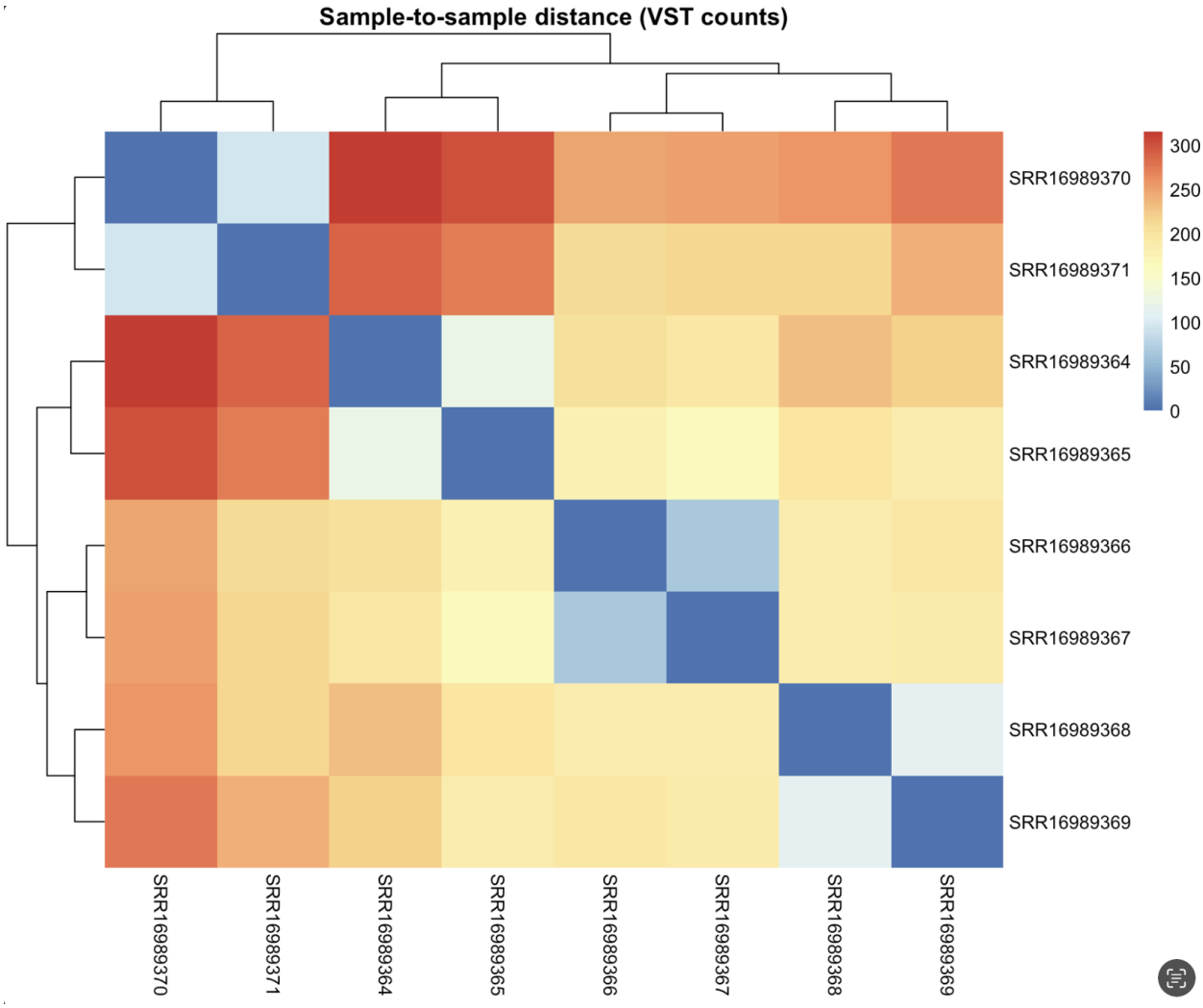


Figure 4: Hierarchical clustering heatmap of sample-to-sample distances calculated from variance-stabilized gene counts.



Reproducibility files

The repository includes Scripts/07_import_salmon.R and Scripts/08_sample_QC.R for expression-matrix generation and sample-level quality assessment, respectively. Differential-expression outputs include session information (R_sessionInfo.txt) and an analysis log (DESeq2_run.log) to document the software environment and major analysis steps. The scripts reference the original local project directory and require path adjustments before execution in another environment.

Differential Expression Analysis

DESeq2 tested expression differences across the four groups.

Statistical approach

Transcript-level Salmon estimates were imported using tximport and analysed with DESeq2. The model used ~ Genotype * Treatment, allowing estimation of genotype effects, treatment effects, and the genotype-by-treatment interaction. Genes with counts ≥10 in at least two samples were retained, leaving 22,242 genes for testing. Significance was defined as adjusted p-value (FDR) <0.05.

Differential-expression summary

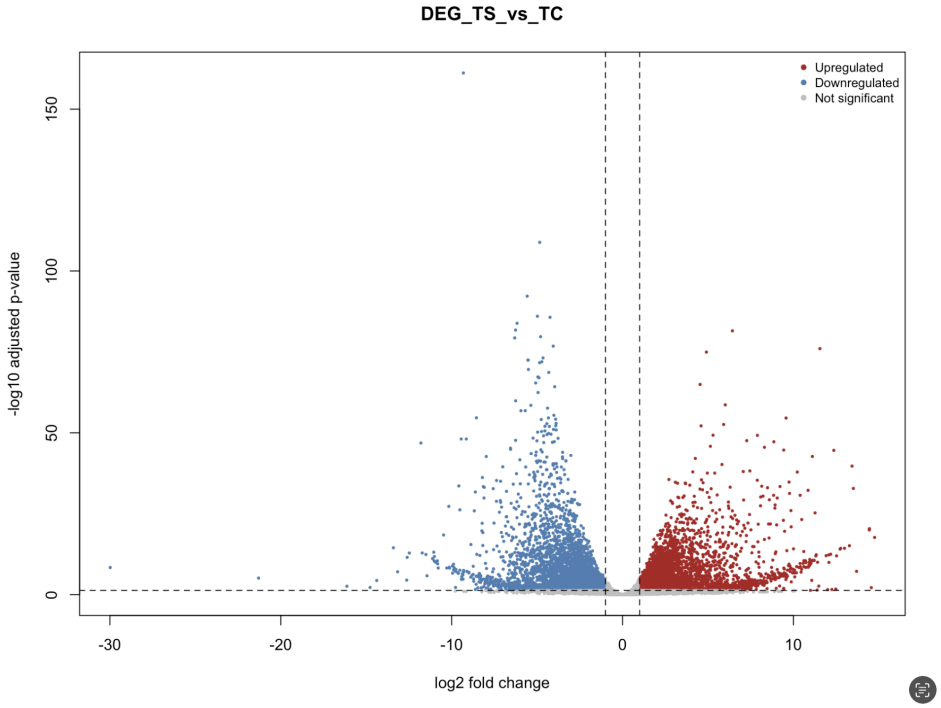
Comparison	FDR <0.05	Up	Down	|log2FC| ≥1
TS_vs_TC	10,154	5,126	5,028	9,301
SS_vs_SC	3,764	1,165	2,599	3,472
TC_vs_SC	5,294	2,367	2,927	4,816
TS_vs_SS	10,916	5,746	5,170	9,913
Genotype × Treatment interaction	6,207	3,398	2,809	6,140

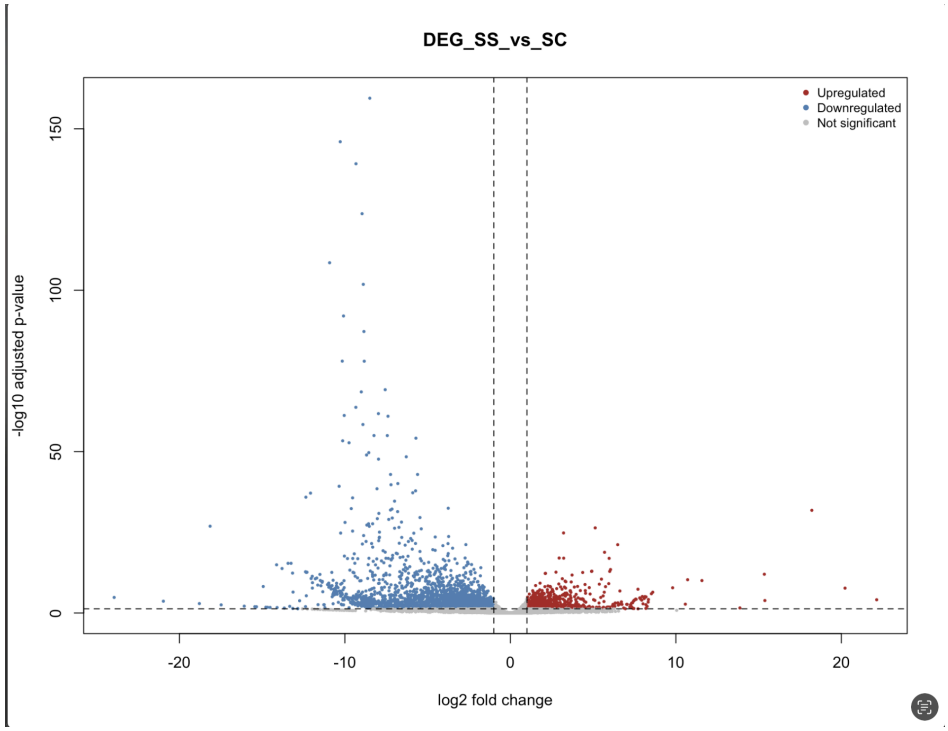
Up/down direction follows the contrast ordering used in the DESeq2 results. For example, a negative log2 fold change in TS_vs_TC indicates lower estimated expression in salt-treated wild/tolerant samples relative to controls, under that contrast definition.

Interpretation

These are outputs of the current exploratory analysis, not proof that every listed gene is a robust salt-response gene. The relatively large DEG sets, two biological replicates per group, and library-type inference discrepancy warrant caution. The analysis can nominate candidate genes for later validation, but mechanisms should not be overclaimed.

Figure 5. Volcano plots showing differential gene expression for (A) salt-treated versus control wild/tolerant barley (TS_vs_TC) and (B) salt-treated versus control cultivated/sensitive barley (SS_vs_SC). The x-axis represents $\log_2$ fold change, and the y-axis represents $-\log_{10}$ adjusted p-value. Red and blue points indicate significantly upregulated and downregulated genes, respectively, while grey points represent genes that do not meet the significance criteria. Dashed lines indicate the fold-change and adjusted p-value thresholds used in the analysis.


DEG_TS_vs_TC
Upregulated
Downregulated
Not significant
-log10 adjusted p-value
0
50
100
150
-30
-20
-10
0
10
log2 fold change


DEG_SS_vs_SC
Upregulated
Downregulated
Not significant
-log10 adjusted p-value
0
50
100
150
-20
-10
0
10
20
log2 fold change

Functional Enrichment and Candidate Interpretation

GO enrichment places selected gene lists into biological context.

Gene-list preparation

For each contrast, significant genes were separated into upregulated and downregulated lists. Background_All_Tested_Genes.txt contained the 22,242 genes tested in differential-expression analysis. Stricter lists were also generated using adjusted p-value <0.05 and log2 fold change ≥+1.5 or ≤−1.5.

GO enrichment workflow

Gene Ontology enrichment was performed using AgriGO SEA with Hordeum vulgare, Genome Version 2.0, user-defined identifiers, and the custom tested-gene background. Fisher's exact test was used with Yekutieli correction; significance was set at 0.05, minimum mapping entries at five, and complete GO terms were selected. Biological Process, Cellular Component, and Molecular Function categories were considered.

Interpretation principles

- Enrichment identifies categories overrepresented in a gene list relative to the chosen background; it does not establish causation.
- Upregulated and downregulated lists should be interpreted separately.
- A comparison with no significant terms is a valid result and should be reported transparently.
- Terms supported by few genes should be treated cautiously given the small replicate count and possible technical confounding.

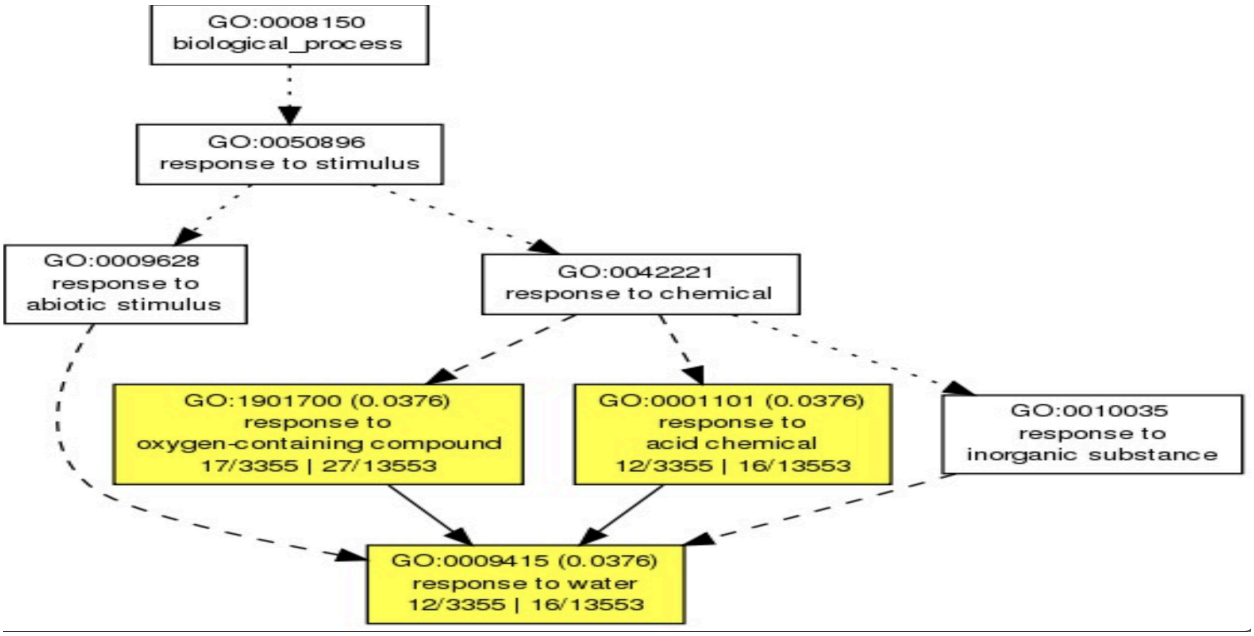


Figure 6. Gene Ontology (GO) biological process enrichment graph for upregulated genes in TS vs. TC.

Candidate follow-up

Candidate genes should be prioritised using adjusted p-value, effect size, expression pattern, annotation confidence, and biological relevance. Any proposed salt-tolerance mechanism should be framed as a hypothesis for follow-up, not a confirmed mechanism.

Limitations and Reproducibility

Transparent reporting is essential for exploratory reanalysis of public data.

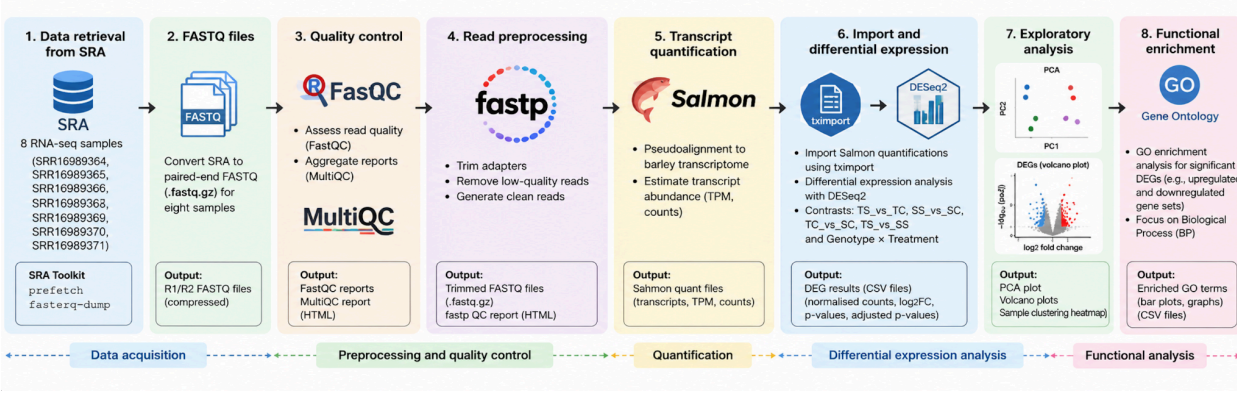
Key limitations

- Replication: each group has two biological replicates, limiting statistical power and stability of estimates.
- Library-type discrepancy: Salmon used automatic inference (libType = A) for all eight samples. Its outputs inferred IU for SRR16989364 and SRR16989365, while the other six were inferred as ISR. Both IU-inferred samples are cultivated controls (SC).
- Protocol mismatch: the study methods have been reported as using a stranded mRNA library protocol, making the IU inference unexpected. The current SRA metadata CSV does not contain library-strandedness or protocol fields, so the discrepancy is not resolved by these metadata files alone.
- Potential confounding: because both IU-inferred samples belong to the same group, technical and biological effects may be difficult to separate. DEG counts and PCA patterns should be treated as exploratory.
- Reference and annotation: the transcript reference and transcript-to-gene mapping influence which features can be quantified and annotated.
- Organism metadata: the SRA organism field lists Hordeum vulgare for all runs, while curated genotype descriptions distinguish wild and cultivated barley. This is retained as a metadata caveat.

Current decision

The existing analysis has been retained without rerunning quantification or DESeq2. This report documents the current outputs and unresolved library-type concern; it does not claim that the discrepancy has been corrected.

Figure 7: Bioinformatics workflow for comparative transcriptomic analysis of salt-stress responses in barley. The workflow begins with retrieval of eight publicly available RNA-seq datasets (SRR16989364–SRR16989371) from the NCBI Sequence Read Archive (SRA), followed by conversion to paired-end FASTQ files using the SRA Toolkit. Raw-read quality was assessed using FastQC, with MultiQC used to aggregate quality-control reports. Read preprocessing was performed using fastp to remove adapter sequences and low-quality reads, generating cleaned FASTQ files. Transcript abundance was then quantified using Salmon against the barley reference transcriptome, producing transcript-level quantification estimates. These estimates were imported into R using tximport to generate gene-level expression matrices for downstream analysis. Principal component analysis (PCA) and sample-to-sample distance clustering were used for exploratory assessment of sample relationships. Differential gene expression was analysed using DESeq2 across five comparisons: salt-treated versus control wild/tolerant barley (TS vs. TC), salt-treated versus control cultivated/sensitive barley (SS vs. SC), genotype differences under control conditions (TC vs. SC), genotype differences under salt treatment (TS vs. SS), and the genotype-by-treatment interaction. Differentially expressed genes (DEGs) were summarised and visualised using volcano plots. Finally, Gene Ontology (GO) enrichment analysis was performed on selected upregulated and downregulated gene sets to identify potentially associated biological processes and functional categories.



Conclusion and Next Steps

A completed computational workflow with clearly stated boundaries for interpretation.

Conclusion

This project established an end-to-end workflow for exploring salt-stress transcriptomic differences in wild/tolerant and cultivated/sensitive barley. Eight paired-end RNA-seq samples were retrieved, quality-checked, preprocessed, and quantified with Salmon. Expression estimates were imported into R, sample relationships were examined using PCA and clustering, and DESeq2 tested four pairwise contrasts plus the genotype-by-treatment interaction. GO enrichment was applied to upregulated and downregulated gene lists.

The analysis produced contrast-specific DEG lists and functional-enrichment outputs that can nominate candidate genes and pathways for further investigation. Because the sample size is small and a library-type inference discrepancy remains unresolved, results are best presented as exploratory. The defensible outcome is the documented workflow, its outputs, and a transparent account of limitations.

References

1. Original research study

Gharaghanipor, N., Arzani, A., Rahimmalek, M., & Ravash, R. (2022). Physiological and transcriptome indicators of salt tolerance in wild and cultivated barley. *Frontiers in Plant Science, 13*, 819282. https://doi.org/10.3389/fpls.2022.819282

2. RNA-seq data source

National Center for Biotechnology Information (NCBI). Sequence Read Archive (SRA). Publicly available barley RNA-seq datasets associated with Gharaghanipor et al. (2022). Run accessions: SRR16989364, SRR16989365, SRR16989366, SRR16989367, SRR16989368, SRR16989369, SRR16989370, and SRR16989371.

Database: https://www.ncbi.nlm.nih.gov/sra

3. Data attribution statement

This project represents an independent computational reanalysis of publicly available RNA-seq datasets originally generated and published by Gharaghanipor et al. (2022). The original authors are credited for the experimental study and sequencing data. The bioinformatics analyses, figures, and interpretations presented in this report document the subsequent reanalysis.